

9.4.34.

Let S be a set of ten integers chosen from 1 through 50. Show that the set contains at least two different (but not necessarily disjoint) subsets of four integers that add up to the same number. (For instance, if the ten numbers are $\{3, 8, 9, 18, 24, 34, 35, 41, 44, 50\}$, the subsets can be taken to be $\{8, 24, 34, 35\}$ and $\{9, 18, 24, 50\}$. The numbers in both of these add up to 101.)

Let $S = \{10 \text{ chosen integers from } 1 \text{ through } 50\}$.

There are different subsets of four integers that can be chosen from S .

$$\binom{10}{4} = \frac{10!}{4!(10-4)!} = \frac{10!}{4!6!} = 210$$

The smallest possible sum of four distinct integers, and the largest possible sum, chosen from 1 through 50:

$$\begin{aligned}1 + 2 + 3 + 4 &= 10, \\47 + 48 + 49 + 50 &= 194.\end{aligned}$$

Since $194 - 10 + 1 = 185$, there are at most 185 possible integer sums. Then:

$$210 \text{ subsets of four integers} > 185 \text{ possible sums.}$$

Therefore, by the pigeonhole principle, there are at least two different subsets of four integers that add up to the same number, which is what was to be shown.

Solution: There are at least two different (but not necessarily disjoint) subsets of four integers that add up to the same number.